Exercise 1

Chapter 1: Introduction to Signals and Systems

Signals

1. Determine whether following signals is periodic/aperiodic and odd/even.
   a. $x(t) = 2\sin 2\pi t$
   b. $x(t) = \frac{1}{|t|}$
   c. $x(t) = \cos(t + \frac{\pi}{4})$

2. Determine the fundamental period of following signals
   a. $x(t) = \cos\frac{\pi}{3}t + \sin\frac{\pi}{4}t$
   b. $x(t) = 2\cos(10t + 1) - \sin(4t - 1)$

3. Determine whether the signals are power signals energy signals
   a. $g(t) = \begin{cases} \sin 2t & 0 \le t \le 2\pi \\ 0 & otherwise \end{cases}$
   b. $h(t) = u(t)$
   c. $f(t) = \sin \omega t$

4. Using property unit impulse function, evaluate following expression
   a. $\int_{-\infty}^{\infty} (t-2)^2 \delta(t-6)\, dt$
   b. $\int_{-\infty}^{\infty} \cos 2\pi t\, \delta(t+2)\, dt$
   c. $\int_{-\infty}^{\infty} e^{-At^2} \delta(t-10)\, dt$, A is a non-zero constant

5. Sketch the following signals
   a. $x(t) = 2u(t) + u(t-1) - 4u(t-2) + 2u(t-3) - u(t-7)$
   b. $y(t) = 2u(t+2) - u(t+1) - 4u(t+1)$
   c. $x(t) = 3(t+3)u(t+3) - 6t\, u(t) + 3(t-3)u(t-3)$.

6. Consider original signal $x(t)$ as in Figure 1

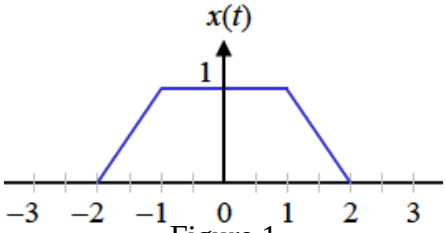


Figure 1

Sketch each of following signal operation on $x(t)$
   a. $x(t-3)$
   b. $x(2t-3)$
   c. $x(-2t-3)$

Systems

7. The following system are described using their input-output relationship. Check the (i) Linearity, (ii) Time-invariant, (iii) Causality of the system
   a. $y(t) = x(t-2)$
   b. $y(t) = x(2t-5)$
   c. $y[n] = n^2 x[n+2]$

8. Two systems are specified by the following input-output relationship.

$$S_1: y[n] = x[n] - 2x[n-1] + x[n-2]$$
$$S_2: y[n] = x[n] + x[n-1] - 2x[n-2]$$

   a) By referring to Figure 2, calculate the input-output relationship for the series configuration of system S_1 and S_2.

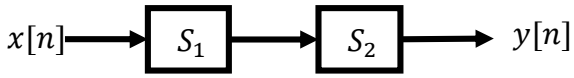


Figure 2

   b) Calculate the input-output relationship for the parallel configuration of system S_1 and S_2 as shown in Figure 3.

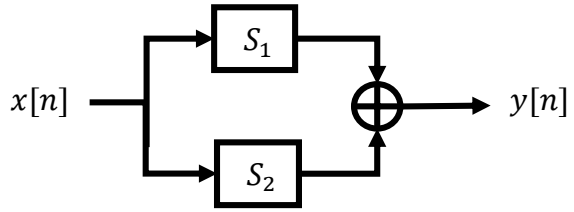
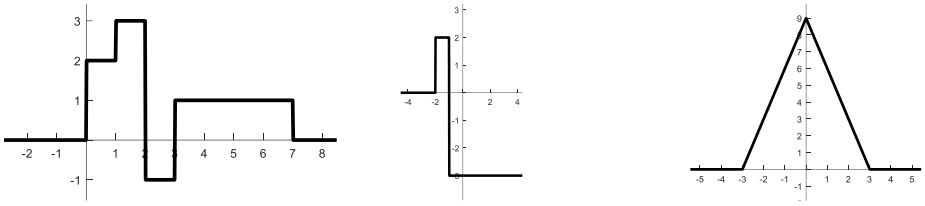


Figure 3

EEE301: Signal and System

Answers

1. a) periodic, odd. b) aperiodic, even. c) periodic, neither even nor odd.
2. a) 24. b) π.
3. a) Energy signal. b) Power signal. c) Power signal.
4. a) 16. b)1. c) e^{-100A}
5. a) b) c)



6. to be shown
7. a) i) Linear, ii) time-invariant, iii) causal
   b) i) Linear, ii) time-varying, iii) non-causal
   c) i) Linear, ii) time-varying, iii) non-causal
8. a) $y[n] = x[n] - x[n-1] - 3x[n-2] + 5x[n-3] - 2x[n-4]$
   b) $y[n] = 2x[n] - x[n-1] - x[n-2]$